

ZONAL'NOYE, Russian Federation

WMO: 299390

Lat: 52.6725N

Lon: 84.9289E

Elev: 225

StdP: 98.65

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -36.4	(c) -32.8	(d) -39.1	(e) 0.1	(f) -36.2	(g) -35.5	(h) 0.1	(i) -32.7	(j) 10.4	(k) -4.6	(l) 9.2	(m) -6.4	(n) 0.7	(o) 60	(p) 0.550

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.7	(c) 30.7	(d) 19.4	(e) 28.9	(f) 18.6	(g) 27.3	(h) 18.1	(i) 21.0	(j) 27.7	(k) 20.0	(l) 26.5	(m) 19.2	(n) 25.4	(o) 2.5	(p) 20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.9	14.1	23.8	17.8	13.1	22.8	16.8	12.3	22.2	62.2	27.9	58.7	26.6	55.8	25.5	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 8.3	(b) 6.6	(c) 5.4	(e) -39.5	(f) 34.0	(g) 4.6	(h) 2.0	(i) -42.9	(j) 35.4	(k) -45.6	(l) 36.6	(m) -48.1	(n) 37.8	(o) -51.5	(p) 39.2		
			DB													
			WB													
				-39.5	23.1	4.3	2.1	-42.6	24.5	-45.1	25.8	-47.6	26.9	-50.7	28.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.8	-17.0	-14.3	-5.7	4.7	12.5	18.1	20.0	17.7	11.5	4.1	-5.8	-13.1		
	DBStd	14.32	8.38	7.99	6.75	5.95	5.17	3.90	3.19	3.66	4.36	5.16	7.81	7.91		
	HDD10.0	3634	838	681	487	174	34	1	0	1	34	193	474	718		
	HDD18.3	5832	1097	914	745	408	192	51	20	56	208	442	723	976		
	CDD10.0	1009	0	0	0	16	111	246	310	239	78	9	0	0		
	CDD18.3	166	0	0	0	0	10	45	72	36	3	0	0	0		
	CDH23.3	1751	0	0	0	13	161	449	642	412	73	1	0	0		
	CDH26.7	488	0	0	0	2	39	116	191	121	19	0	0	0		
(14) Wind	WSAvg	2.1	1.9	2.0	2.2	2.6	2.4	2.0	1.7	1.7	1.8	2.3	2.4	2.3		
(15) Precipitation	PrecAvg	534	32	27	24	36	50	54	59	64	46	50	49	40		
	PrecMax	746	96	79	109	108	114	114	153	148	114	101	137	90		
	PrecMin	330	4	3	1	6	8	12	6	17	12	10	5	6		
	PrecStd	104	19	17	20	22	24	26	37	32	27	24	30	20		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.6	2.7	9.8	25.1	30.5	31.8	32.9	33.3	29.4	21.6	12.1	3.3		
		MCWB	1.0	0.3	5.3	13.6	16.8	19.0	20.8	20.1	16.7	11.6	7.0	0.8		
	2%	DB	-0.1	0.7	6.9	20.8	27.5	29.5	30.6	29.7	25.1	17.7	8.6	1.2		
		MCWB	-1.4	-1.1	3.3	11.5	15.4	18.3	20.3	19.3	15.2	10.2	5.0	-0.6		
	5%	DB	-2.6	-1.0	5.0	17.5	25.0	27.9	28.7	27.5	22.2	14.8	5.6	-0.5		
		MCWB	-3.8	-2.6	2.1	9.9	14.1	18.2	19.7	18.3	14.0	9.0	3.0	-1.9		
	10%	DB	-5.1	-3.1	3.0	14.3	22.2	26.0	27.0	25.5	19.6	12.0	3.1	-2.3		
		MCWB	-6.2	-4.4	0.8	8.3	13.2	17.4	19.1	17.5	12.8	7.6	1.3	-3.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.1	0.6	5.9	14.1	18.5	21.0	23.2	21.8	17.9	12.4	7.7	1.2		
		MCDB	2.5	2.0	9.4	23.1	27.7	28.0	29.6	29.7	25.7	19.5	10.7	2.6		
	2%	WB	-1.3	-1.0	3.8	12.1	16.6	19.9	21.6	20.2	16.0	10.7	5.0	-0.1		
		MCDB	-0.1	0.6	6.5	19.0	24.6	26.6	27.7	27.4	23.3	16.4	8.0	1.1		
	5%	WB	-3.7	-2.6	2.3	10.3	15.3	19.0	20.6	19.1	14.8	9.4	3.3	-2.0		
		MCDB	-2.5	-1.2	4.6	16.7	22.4	25.7	26.6	26.0	21.1	14.4	5.6	-0.8		
	10%	WB	-6.1	-4.4	0.9	8.8	14.0	18.2	19.7	18.1	13.5	7.9	1.4	-3.5		
		MCDB	-5.1	-3.1	2.9	13.9	20.5	24.6	25.6	24.3	18.7	11.7	3.0	-2.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.2	11.6	11.1	11.1	12.1	11.2	10.7	11.7	11.9	9.7	8.0	9.0		
		MCDBR	9.4	10.6	10.7	17.0	16.3	14.4	13.6	15.3	17.0	15.7	9.8	9.0		
		MCWBR	8.3	9.1	7.7	9.1	7.1	6.0	5.6	6.4	8.1	8.8	6.7	7.9		
		MCDBR	9.5	10.4	9.9	15.5	13.4	12.1	11.7	13.0	14.9	14.5	9.3	9.2		
	5% WB	MCWBR	8.6	9.1	7.6	8.7	6.6	5.7	5.4	6.0	7.7	8.5	6.6	8.5		
(40) Clear-Sky Solar Irradiance	taub		0.299	0.338	0.391	0.424	0.416	0.412	0.436	0.412	0.373	0.363	0.346	0.310		
	taud		2.011	1.944	1.906	2.076	2.176	2.272	2.212	2.255	2.310	2.213	2.033	1.957		
	Ebn at Noon		638	726	767	803	833	841	811	810	798	707	566	529		
	Edh at Noon		80	117	151	145	138	127	133	120	101	90	79	71		
(44) All-Sky Solar Radiation	RadAvg		1.15	2.23	3.67	4.66	5.57	6.31	5.74	4.93	3.50	1.89	1.02	0.81		
	RadStd		0.10	0.13	0.35	0.31	0.57	0.42	0.37	0.32	0.32	0.22	0.13	0.10		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page